Eppala Mohana

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.52
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2023	99.20%
Narayana High School	Class X (Telangana State Board of Secondary Education)	2021	10.00%

WORK EXPERIENCE

American Express -- Data Steward Intern, Global Risk & Compliance

- Performed large-scale data profiling, anomaly detection, and root-cause analysis across **500+** customer and transaction data elements, identifying data quality issues and governance gaps.
- Developed automated data quality assessment solutions using SQL and analytical techniques, streamlining validation across multiple data quality dimensions.
- Engineered **DataAssist, an LLM-powered agentic chat interface** for interactive metadata exploration and enterprise knowledge retrieval, reducing analysis time by **95%.**
- DataAssist is a **tool-augmented LLM architecture** that enables controlled access to data profiling results without directly exposing sensitive enterprise data, supporting scalable analysis across datasets and markets.

Undergraduate Research Intern - Machine Learning

- **Audio Signal Enhancement using Deep Neural Network -- DeepFilterNet**
- Implemented **DeepFilterNet2**, an advanced real-time speech enhancement model for denoising noisy audio signals
- DeepFilterNet2 combines an **encoder-decoder U-Net**, deep filtering, spectral masking, and multi-band processing for effective noise suppression
- Optimized the model for causal, low-latency inference on edge devices while maintaining high speech quality

PROJECTS

Event-Driven User-Level Thread Scheduler & Redis Compatible Datastore | C++, Linux, TCP/IP

- Built a **Redis-compatible in-memory datastore** from scratch in C++20 with an event driven architecture and user level thread scheduler for high-concurrency workloads on a single CPU core.
- Implemented **non-blocking TCP connection handling and user-level scheduling**, supporting **10,000 concurrent** clients while minimizing kernel thread-scheduling overhead.
- Benchmarked with redis-benchmark, achieving **~90K requests/sec at 1K concurrent clients** and **63K SET/ 71K GET requests/sec at 10K clients (C10K)**.

ForgeBuild -- Generic Incremental Build System | C++20, CMake

- Architected a configuration-driven build engine leveraging **Directed Acyclic Graphs (DAGs)**, utilizing **Kahn's algorithm** for topological sorting and cycle detection to reliably orchestrate complex execution pipelines.
- Engineered a **thread-safe, parallel task scheduler** using **std::mutex** and condition variables, maximizing CPU utilization by concurrently executing independent subgraphs while ensuring deterministic failure propagation.
- Optimized end-to-end execution times by implementing an incremental build planner; utilized **persistent JSON state caching**, **I/O metadata tracking**, and **command hashing** to accurately propagate cache invalidations and eliminate redundant work.

VWAP Execution Simulator | Python

- Developed a **Volume Weighted Average Pricing** (VWAP) execution simulator using 1-minute historical market data, slicing parent orders according to intraday volume distributions and configurable execution schedules.
- Implemented realistic execution modeling with **bid-ask spread, temporary market impact and partial fills**,

computing metrics including VWAP slippage, implementation shortfall, participation rate and average execution price.
- Performed **100+** simulation runs under varying liquidity, spread and market impact assumptions to analyze execution robustness and transaction cost sensitivity.

SCHOLASTIC ACHIEVEMENTS

- Secured National Rank of **93** in Physics Olympiad by SIMO Education
- Secured rank **439** in TS Eamcet (State Engineering Entrance Exam) among 3.3 lakh candidates

SKILLS

Programming Languages : C | C++ | Python | SQL | MATLAB

Technical Skills : Data Structures and Algorithms | OS | OOPS | Machine Learning

Tools and Software : Google Bigquery | Microsoft PowerBI | Power Query| CMake | Git | Linux | Jupyter

RELEVANT COURSES

Mathematics : Probability and Random Processes | Matrix Theory | Introduction to Statistics | Calculus | Differential Equations | Vector Calculus | Intro to Metric Spaces

Coding & ML : Data Structures and Algorithms | Artificial Intelligence | Introduction to Programming(C) | Machine Learning | Deep Learning | Reinforcement Learning

POSITIONS OF RESPONSIBILITY AND EXTRACURRICULAR

- Public Relation Coordinator for Milan (Annual Event of IITH)
- Buddy at Sunshine (Counselling cell of IITH)
- Member of National Cadet Corps (NCC)